



Maximising wireless charging efficiency with multiple transmitters



Wireless charging has already found its way in many consumer electronics and medical applications. Maximising efficiency when using multiple transmitters could extend its use to industrial robotics and electric vehicles

Wireless power transfer has proven to be quite useful in electronic devices. Although single-transmitter systems can provide maximum efficiency for wireless transfer of power, achieving the same with multiple transmitter coils is challenging. Scientists at the Incheon National University, South Korea have now devised an effective control strategy that can maximise efficiency in multi-transmitter wireless charging.

To accomplish this, relationships between many variables in the problem were established, such as the connection between the degree of coupling of each transformer to the receiver, its 'perceived' or 'reflected' impedance from the receiver, and the optimal current that should be fed. This allowed the implementation of a novel, maximally efficient, and relatively simpler method for multi-transmitter wireless charging.

Soft and sensitive robotic finger design to provide accurate grip

One of the main challenges in the design of soft robotic grippers is the integration of traditional sensors onto a robot's fingers. Ideally, a soft gripper should have proprioception—a sense of its own movements and position for safely executing varied tasks. However, traditional sensors are rigid and compromise the mechanical characteristics of the soft parts.

To overcome these limitations, scientists at Ritsumeikan University, Japan have successfully used multi-material 3D printing technology to fabricate soft robotic fingers with a built-in proprioception sensor consisting of vacuum and piezoelectric technology. This design strategy offers numerous advantages and

represents a large step towards developing safer and more capable soft robots.

Get a Grip: 3D Printed Sensitive Fingers for Soft Robots

Human-like grasping is a top goal in robotics...

...but integrating traditional sensors into soft grippers is difficult!

Hurdles

- Undesired mechanical characteristics
- Requires power supply

Introducing...

3D-printed soft robotic finger with multifunctional sensor

- Multimaterial 3D printing
- Easy fabrication
- Inflation chamber
- Provides curvature
- Vacuum jamming layers
- Tunable stiffness
- Piezoelectric layer
- Senses curvature and stiffness
- No energy source required!

Soft robotic gripper!

Better interaction between robots and their environment

Applications in medicine and elderly care